Hosein Gharakhani

AI / LLM Engineer

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PROFILE

AI / LLM Engineer focused on building practical applications with large language models, RAG, tool calling and fine-tuning. Hands-on experience with OpenAI APIs, Hugging Face Transformers, ChromaDB, Gradio, SQLite and QLoRA. Built end-to-end projects covering retrieval-augmented generation, voice applications, structured data pipelines and LLM fine-tuning.

CORE SKILLS

- AI / LLM: LLMs, Generative AI, RAG, Embeddings, Tool Calling, Prompt Engineering, LoRA / QLoRA, Fine-tuning
- Frameworks & APIs: Python, OpenAI API, Hugging Face Transformers, Gradio, PyTorch
- Data & ML: ChromaDB, SQLite, Pydantic, NumPy, XGBoost, Random Forest, Data Preprocessing, Model Evaluation
- Other: Whisper, Text-to-Speech, Image Generation, Git / GitHub, CUDA

SELECTED PROJECTS

Used Car Price Predictor — LLM Fine-Tuning with QLoRA

Fine-tuned Meta Llama 3.2 3B for used-car price estimation using supervised fine-tuning and QLoRA. Built a Pydantic-based data pipeline, transformed structured vehicle records into training prompts, trained with 4-bit NF4 quantization on a T4 GPU, and evaluated predicted vs. actual prices.
Tech: Python · Llama 3.2 3B · QLoRA · Hugging Face · Pydantic · PyTorch · 4-bit NF4
github.com/HoseinGharakhani/used-car-price-prediction

TechShop RAG

Built a Persian RAG assistant for a technology store. Prepared Markdown knowledge sources, created embeddings, stored vectors in ChromaDB, retrieved relevant context and generated answers through an LLM-powered Gradio interface.
Tech: Python · OpenAI API · ChromaDB · Embeddings · Gradio
github.com/HoseinGharakhani/techshop-rag

AI Pizza Shop Assistant

Developed a conversational pizza-shop assistant connected to SQLite with tool calling for product pricing and order placement. Added voice interaction, text-to-speech and image-generation capabilities.
Tech: Python · GPT-4o-mini · OpenAI API · SQLite · Gradio · Tool Calling · TTS
github.com/HoseinGharakhani/ai-pizza-shop-assistant

Universal Code Forge

Built a Gradio application for generating and converting code between programming languages using LLMs, with support for multiple code-focused and general-purpose models.
Tech: Python · LLM · OpenAI API · Gradio · python-dotenv
github.com/HoseinGharakhani/universal-code-forge

Whisper Speech-to-Text

Implemented an English speech-to-text pipeline using OpenAI Whisper with GPU acceleration and timestamped transcription.
Tech: Python · Whisper · Hugging Face Transformers · PyTorch · CUDA
github.com/HoseinGharakhani/whisper-speech-to-text

EDUCATION

Self-directed training in AI / LLM Engineering — practical project-based learning

TARGET ROLE

AI / LLM Engineering roles involving LLM applications, RAG, agents, tool calling, evaluation, fine-tuning and AI automation.